

R BHARAT SINGH

(R&D Software Engineer)

+91-8296495038

rbhartsingh@gmail.com

Objective

To work in a challenging environment, where I could implement my technical skills to innovative, prove my caliber and adapt myself to the emerging technologies.

Professional Summary

Having 4.6 years of experienced as R&D Software Engineer in the Telecom Domain and having experience in windows device driver including Design and Development of Networking Protocols using C & C++ language in Windows/Linux environment. Domain knowledge in 5G/LTE/5G NR/IP networking and TCP/IP.

Key Skill Areas

- Worked as Software engineer in various domains of a WLAN & Networking Protocols and Windows Device Drivers.
- Expertise in C/C++ programming and software in Linux/Windows environments.
- Having basic knowledge of networking protocols.
- Worked on IPC and having good understanding in multi-threading concepts.
- Excellent diagnostics, debugging and troubleshooting experience on Linux environments (GDB, Valgrind etc), along with ability to identify product improvements.
- Hands on Experience with Wireshark, Omni peek, Xshell, MobaXterm and Tera-term, Jira and GitHub.

ACADEMICS

2018 - 2019 – **Embedded Course** from **Vector Institute, Bangalore, India.**

2015 - 2018 – B. Tech (aggregate 66%) in **Electronics and Communication Engineering** from Dr.A.P.J Abdul Kalam Technical University.

2012 - 2015 – BSC (64%) in Computer Science from Dr Bhim Rao Ambedkar University.

2011 - 2012 – Higher Secondary (63%) from CBSE Board, India.

Technical Experience

- Programming Languages – C/C++, Data-structure and Linux.
- Domain knowledge in 4G/5G/LTE/SRAN/IP and networking Protocols.
- Excellent Pronto debugging skills and Design and implementation of BTS Transport features in 5G/LTE/SRAN.
- Packet Analyzer – Omni peek, Tera-Term, Wireshark, Tshark, Ixia and Tcpdump.
- Wireless Technology- IEEE 802.11 WLAN Protocol and Standards.

Employment Details

- Nokia Solutions and Networks Pvt Ltd, Bangalore (April-2022 to Dec-2023).
- Intelligraphics Technology India Pvt Ltd, Bangalore (April-2019 to Feb-2022).

Work Experience in Nokia Solutions & Networks

Organization: 1 Year and 8 months of experience in Telecom Domain including Design & Development of Networking Protocols using C/C++ language in Linux environment as R&D Software Engineer at **Nokia Solutions and Networks Pvt Ltd**, Bangalore.

PRODUCT: - ASOE, ASIL, ASIM, Flexi Zone Micro & Macro.

PROJECT (Features): -

- 1- CB009344: FZ OPER parity: ITSAR & SCAS compliant user event records (Flexizone Micro feature related to **LTE/4G**).
- 2- TRSEE-31311 - Support to increase in TNL instances and Mocs.
- 3- CB010461-SR FZM DPD and IPsec SA reestablishment mechanism improvement.
- 4- TRSEE-36969 - ASOF Ethernet OAM + ASOF Traffic Capture.

Roles/Responsibilities

- Participated to develop the code for all these features related to LTE/4G/5G.
- Fixing the issue in current development and all Pronto related to these features.
- Write the test cases (MT) and test plan for SCT testing.
- Work on Linux based environment.

Work Experience in Intelligraphics Technology

Organization: 2 Years and 11 months of experience in Windows Device Driver Development as Software Engineer at **Intelligraphics Technology India Pvt Ltd**, Bangalore.

PRODUCT: - WiOpsy 802.11ac USB Windows WLAN Sniffer (L1/L2), WEC7 USB Driver, 802.11acSniffer/Injector for QCA9378/QCA6174, TeckAR9592 windows drivers, WDI drivers.

Project I: WiOpsy WLAN Sniffer

WiOpsy is used to perform the capture of the Wireless traffic and analyze the data, to debug and diagnose any of the Wi-Fi deployment field issues. It is a Wi-Fi Sniffer that monitor and capture Wi-Fi Frames in a particular channel (Frequency). It is included some other features also like Injector, Arial View, 5 & 10 MHz support and Spectral Scan and more.

Team size 5
Language Used C, Data-structure and C++
(UI)
Role Software Engineer
Environment Windows
Tools used Visual Studio, QT , Tera Term, Wireshark, and InstallShield Wizard.

Roles/Responsibilities

- Participated to develop the code for QCA9378 WiOpsy Driver.
- Fixing the issue in current development.
- Support in testing the WiOpsy WLAN Sniffer/Injector and live capture and offline analysis.
- Support in implementation of UI and Application development.

Project II: WEC7 USB Driver

Windows Embedded Compact 7 (WEC 7) is a new operating system that share and manage content across Network Device. It is supported IEEE 802.11 a/b/g/n and Qualcomm Atheros USB and PCIs Wi-Fi adapters.

Team size 3
Language Used C and Data Structure
Role Software Engineer
Environment Windows Embedded Compact
Tools used Visual Studio, QT, Tera Term and InstallShield.

Roles/Responsibilities

- Main role to develop the code for QCA9377 (WEC7) USB Device Driver.
- Support in testing the Iperf and Wi-Fi connections with different adapters.
- Fixing the issue in current development.

Personal Profile

Name R Bharat Singh
Date of Birth 19-06-1994
Sex Male

Marital Status	Married
Nationality	Indian
Languages Known	English, Hindi

Declaration: - I am also confident of my ability to work in a team. I hereby declare that the information furnished above is true to the best of my knowledge.

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